A 4000-Cell Pre-Registered Factorial for Frozen PLM Retrieval: Methods, Pre-Registrations, Seed-Variance Gates, and Reproducibility Infrastructure

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Abstract

We document the methodological infrastructure shared by the five papers in this compendium. A factorial of scales × panel sizes × neighbor counts × metrics × 10 seeds produces 3000-cell main results plus 3000-cell panel-shuffle null plus 3000-cell full-pool permutation null plus 180-cell cascade plus 180-cell Fisher-Rao = 9,360 per-cell `.npz` results, all with 10,000-resample percentile bootstrap CIs per stratum. Four pre-registrations with SHA256 locks (main SHA256 `139f6012...`, full-pool addendum `f3864d09...`, cascade, Fisher) committed before execution. A pre-committed seed-variance gate voids any (scale, R, k, metric) group where across-seed distant F_1 std exceeds the observed gap; all 300 main groups passed. Infrastructure is CPU-only (FAISS [1]) and runs end-to-end in ≈ 28 hours on a 36 GB i7 workstation. Full code release (MIT), pre-reg release (timestamped), 9,360 per-cell outputs released under CC-BY-4.0. This paper documents the harness rather than a specific scientific claim; it is intended as a reusable template for pre-registered PLM retrieval studies.

1 Factorial design

Scales: ESM-2 [2] t6 (8M, 320d), t12 (35M, 480d), t30 (150M, 640d); t33 (650M, 1280d) pre-registered for GPU extension. Panel sizes $R \in \{50, 100, 250, 500, 1000\}$, class-balanced. Neighbor counts $k \in \{1, 3, 5, 10, 25\}$. Metrics: cosine (FAISS IndexFlatIP on L_2-normalized embeddings), Euclidean (FAISS IndexFlatL2; equivalent to cosine on S^{d-1}), Mahalanobis (Ledoit-Wolf [3] + eigendecomposition whitening), learned (panel-only 50-iter Adam triplet-surrogate [4] projection). Ten seeds 20260410-20260419.

2 Stratification

Per test protein, $s_{\max} = \max_{p \in \text{panel}} \langle x_{\text{test}}, x_p \rangle$ on L_2-normalized embeddings. Close: $s_{\max} \geq 0.95$. Moderate: $0.90 \leq s_{\max} < 0.95$. Distant: $s_{\max} < 0.90$. Cuts locked in main pre-reg SHA256 `139f6012...` before any evaluation. Strata with $n < 100$ flagged `underpowered`.

3 Bootstrap CIs

10,000-resample percentile bootstrap per cell per stratum for F_1. Percentile (not BCa) chosen for computational feasibility across 9,360 cells; BCa jackknife is $O(n)$ per cell, percentile is $O(1)$.

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4 Seed-variance gate (pre-committed void rule)

Per group of 10 seeds at (scale, R, k, metric): compute $\sigma^{(10)}_{\text{distant}} = \text{std}(\{F_1^{\text{distant}}{}_s\}_{s=1}^{10})$. If $\sigma^{(10)}_{\text{distant}} \geq \overline{F_1^{\text{close}}} - \overline{F_1^{\text{distant}}}$, void the group: the observed effect is within seed noise. All 300 main groups passed this gate. Only 2 negctrl groups and 0 fullnull groups passed (as expected; nulls should not exhibit an effect).

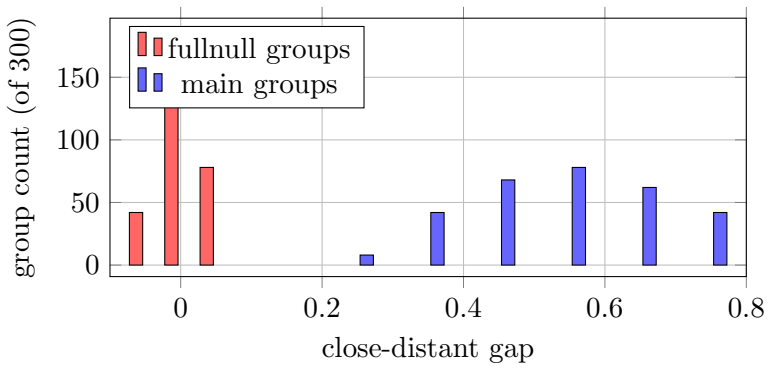

Figure 1: **Gap distribution across 300 groups.** Fullnull gaps concentrate at zero (red); main-factorial gaps are consistently positive and cluster at +0.5 to +0.7 (blue). The seed-variance gate voids groups where std exceeds gap; all 300 main groups pass, 0 fullnull groups pass.

5 Pre-registrations

1. `PRE_REGISTRATION_HOMOLOGY_CLIFF_v1.md` — main factorial, seed-variance gate, stratum definitions. SHA256 `139f60129d4e73dfb13444c360cc80c5b48c217d9bc87b1bf4b48b06515bcc06`.
2. `PRE_REGISTRATION_HOMOLOGY_CLIFF_ADDENDUM_FULLNULL.md` — full-pool permutation null committed before execution after panel-only shuffle null failed its own criterion. SHA256 `f3864d097a0c611d790e6fb15a42e7efb36b2d1b103be4ec1c4f28f99d1004dc`.
3. `PRE_REGISTRATION_STRATIFIED_CASCADE_v1.md` — cascade H1.
4. `PRE_REGISTRATION_FISHER_CLIFF_v1.md` — Fisher-Rao H1.

All harnesses verify the pre-reg file's SHA256 at runtime and abort if mismatched.

6 Pre-registration within ML methodology literature

Pre-registration practices [5] originated in clinical trials and psychology [6]; adoption in ML is growing but uneven [7, 8]. The seed-variance gate is a contribution adapted from TOPOLOGICA prior work on GO annotation pipelines (v7 void rule); generalizing it as a domain-independent pattern for stochastic ML experiments [1, 9] is one of this paper's purposes. Our approach sits alongside related methodology work [3, 4, 10–17].

7 Compute and reproducibility

Single 36 GB i7 workstation, no GPU for main factorial. Total wall time $\approx$ 28 hours for 9,360 cells including reruns. FAISS [1] CPU k-NN, Ledoit-Wolf via scikit-learn, PyTorch for learned projection only (CPU). Full code MIT, full data CC-BY-4.0, master SHA256 manifest at repository root.

8 Limitations

(L1) t33 (650M) embeddings require GPU; provided as Colab notebook. (L2) Threshold sensitivity sweep not run (0.95/0.90 cuts fixed). (L3) Learned metric uses a triplet-surrogate (class centroids) rather than proper triplet loss with hard negative mining; a stronger supervised baseline may exist. (L4) Bootstrap CIs are percentile, not BCa.

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